STACKS, DEFORMATION THEORY: LEMMA-INF-OBS-MAP-DEFO-COMPLEX

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 5752-5768 of sources/stacks-defos/pinned-original-source.tex; proof span is lines 5770-5800, the complete author proof (29 body lines).

Editorial note. Navigation stubs: 1 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 29 of queue_defos.json (pb=1041, stmt 5752-5768, proof 5770-5800).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 0149: sources/stacks-defos/pinned-original-source.tex, source lines 5800–5800. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $\mathcal{C}$ be a site. Let $\mathcal{O} \to \mathcal{O}_0$ be a surjection of sheaves of rings whose kernel is an ideal sheaf $\mathcal{I}$ of square zero. Let $K, L \in D^-(\mathcal{O})$. Set $K_0 = K \otimes^{\mathbf{L}}_{\mathcal{O}} \mathcal{O}_0$ and $L_0 = L \otimes^{\mathbf{L}}_{\mathcal{O}} \mathcal{O}_0$ in $D^-(\mathcal{O}_0)$. Given $\alpha_0 : K_0 \to L_0$ in $D(\mathcal{O}_0)$ there is a canonical element*

$$o(\alpha_0) \in \mathrm{Ext}^1_{\mathcal{O}_0}(K_0, L_0 \otimes^{\mathbf{L}}_{\mathcal{O}_0} \mathcal{I})$$

whose vanishing is necessary and sufficient for the existence of a map $\alpha : K \to L$ in $D(\mathcal{O})$ with $\alpha_0 = \alpha \otimes^{\mathbf{L}}_{\mathcal{O}} id$.

Proof. Finding $\alpha : K \to L$ lifing α_0 is the same as finding $\alpha : K \to L$ such that the composition $K \xrightarrow{\alpha} L \to L_0$ is equal to the composition $K \to K_0 \xrightarrow{\alpha_0} L_0$. The short exact sequence $0 \to \mathcal{I} \to \mathcal{O} \to \mathcal{O}_0 \to 0$ gives rise to a canonical distinguished triangle

$$L \otimes^{\mathbf{L}}_{\mathcal{O}} \mathcal{I} \to L \to L_0 \to (L \otimes^{\mathbf{L}}_{\mathcal{O}} \mathcal{I})[1]$$

in $D(\mathcal{O})$. By Derived Categories, Lemma 0149 the composition

$$K \to K_0 \xrightarrow{\alpha_0} L_0 \to (L \otimes^{\mathbf{L}}_{\mathcal{O}} \mathcal{I})[1]$$

is zero if and only if we can find $\alpha : K \to L$ lifting α_0. The composition is an element in

$$\mathrm{Hom}_{D(\mathcal{O})}(K, (L \otimes^{\mathbf{L}}_{\mathcal{O}} \mathcal{I})[1]) = \mathrm{Hom}_{D(\mathcal{O}_0)}(K_0, (L \otimes^{\mathbf{L}}_{\mathcal{O}} \mathcal{I})[1]) = \mathrm{Ext}^1_{\mathcal{O}_0}(K_0, L_0 \otimes^{\mathbf{L}}_{\mathcal{O}_0} \mathcal{I})$$

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by adjunction. □